

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 10%

QUESTIONS: 2

Duration : 45 Minutes
Total Marks:20

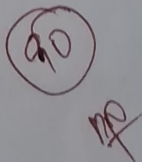
Annexure A

Descriptive Written Test

Code of Conduct:

I Sanjana Sree.P (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Unit 1 : Assessment Tool 3

Descriptive written Test

Course Code : CSA710 Artificial
Intelligence

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1). Smart Home Automation System and Intelligent Agents

Introduction :

A Smart home automation system manages lighting, temperature and security using inputs such as motion sensors, temperature sensors and time of day.

Intelligent agents help the system make decisions automatically.

Types of Intelligent Agents :

1). Simple Reflex Agent :

(*) Acts only on current sensor inputs using predefined rules.

(*) Eg: If motion is detected $\rightarrow$ Turn on lights.

If temperature $> 30^{\circ}\text{C}$ $\rightarrow$ Turn on air conditioner.

(*) Advantages : Fast response and easy implementation.

(*) Limitations : Cannot remember past events or learn user preferences.

2. Model-Based Agent:

- (*) Maintains an internal model of the environment.
- (*) Keeps track of room occupancy, temperature history and device status.
- (*) Eg: If no movement is detected for a long period, lights are turned off.
- (*) Advantages: Better decision-making using past information.
- (*) Limitations: Requires more memory and processing power.

3. Local-Based Agent:

- (*) Takes actions to achieve specific goals.
- (*) Eg: Maintain comfortable room temperature.
Ensure home security.
Reduce electricity consumption.
- (*) Advantages: flexible and intelligent.
- (*) Limitations: Planning can be computationally expensive.

4. Utility-Based Agent:

- (*) Chooses actions that maximise overall utility.
- (*) Eg: Selects an optimal temperature that balances user comfort and power consumption.
- (*) Advantages: Produces the most beneficial outcome.

2). Robot Navigation in an unknown Maze

Introduction :

A robot is placed in an unknown maze and must find the exit without prior knowledge of the maze layout. Search algorithms help the robot explore and discover the correct path.

Uniform Cost Search (UCS) :

Working :

- (*) UCS expands the node with the lowest path cost.
- (*) It guarantees the least-cost path to the goal.

Example :

If moving through different paths has different costs, UCS always selects the path with minimum cumulative cost.

Advantages :

- (*) Complete (find a solution if one exists)
- (*) Optimal (finds minimum-cost path).

Disadvantages :

- (*) Requires large memory.
- (*) Can be slow for large mazes.

Iterative - Deepening Search (IDS):

Working:

- (*) Repeatedly performs Depth-limited Search.
- (*) Increases depth limit gradually until the goal is found.

Advantages:

- (*) Requires very little memory.
- (*) Complete for finite search space.
- (*) Suitable when solution depth is unknown.

Disadvantages:

- (*) Repeats exploration of nodes.
- (*) Not optimal when path costs vary.

Relation to Intelligent Agents:

Simple Reflex Agent:

- (*) Acts only on current observation.
- (*) May get trapped in loops.

Model-Based Agent:

- (*) Maintains a map of explored locations.
- (*) Avoids revisiting states unnecessarily.

Smart Traffic management System and Intelligent Agents :

Introduction :

A Smart traffic management controls traffic signals using vehicle density, pedestrian movement and weather conditions. The aim is to reduce congestion and improve traffic flow.

Types of Intelligent Agents :

1). Simple Reflex Agent :

(*) Uses current sensor inputs only.

(*) Eg : If vehicle count exceeds threshold $\rightarrow$ Extend green signal time.

(*) Advantage : Fast Response.

(*) Limitation : Cannot predict future traffic conditions.

2). Model - Based Agent :

(*) Maintain information about traffic conditions.

(*) Tracks traffic density and previous signal states.

(*) Eg : Remembers congestion levels at different intersections.

(*) Advantage : Better decisions using historical data.

3). Goal-Based Agent :

(*) works toward goal such as :

→ Reduce waiting time

→ Improve traffic flow.

→ Increase pedestrian safety.

(*) Evaluates alternative signal timings before acting.

4). Utility-Based Agent :

(*) Maximizes overall traffic efficiency.

(*) Considers :

→ Travel time

→ Congestion

→ Fuel consumption

→ Safety.

(*) Chooses actions with the highest utility value.

